

Through-focal inversion of LEEM intensities with the Fourier-optics kernel

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Abstract

We present a through-focal inversion of low-energy electron microscopy (LEEM) intensities for the object exit wave $\psi_0 = \sigma e^{i\phi}$, based on the Fourier-optics (FO) kernel of Yu, Lau and Altman [11]. Prior FO studies compared predicted and measured contrast for prescribed objects; they did not recover ψ_0 . We invert that map in two stages: regularized multi-defocus inversion on the contrast-transfer slice $R_{\text{FO}}(\mathbf{q}, 0, \Delta z)$, which fills zeros of a single-defocus contrast-transfer function (CTF), followed by optional Gauss–Newton iteration on the full bilinear kernel, omitted when the linearized residual lies at the noise floor ($\max|\phi| \lesssim 0.3$). A one-dimensional sinusoid reduces the second stage to Jacobi–Anger fitting of the aperture-truncated Bessel ladder. Uniqueness of the regularized Fourier-diagonal estimator and its bias–noise identity are machine-checked in Lean 4 against that kernel. This work extends quantitative evaluation of LEEM contrast from forward matching of a focal series to recovery of the object that produced it.

1 Introduction

Image interpretation in LEEM [1, 2, 3, 4] requires a quantitative account of how the cathode lens, contrast aperture, source size and energy spread map the object exit wave to the recorded intensity. Aberration-corrected instruments [5, 6] make that map sharper, but they do not remove the distinction between two standard calculations. The contrast-transfer function (CTF) evaluates the inverse transforms of ΨH and $\Psi^* H^*$ separately and is computationally efficient [8, 9]. Fourier optics (FO) retains inseparable mixing of spatial-frequency pairs $(\mathbf{q}, \mathbf{q}')$ under partial coherence [7, 10, 11]. The mathematically rigorous FO calculation gains an advantage under conditions, particularly defocus, in which partial coherence compromises the validity of the CTF intensity [11]. Yu, Locatelli and Altman [10] showed that this distinction is not academic: CTF modelling fails for strongly rippled graphene, whereas FO reproduces the measured through-focal contrast.

Those studies prescribe an object, evaluate the forward map, and compare predicted and measured contrast versus defocus. They do not invert. The only demonstrated 2D experimental LEEM exit-wave reconstruction is the linear focal-series restoration of Duden *et al.* [12], in the tradition of Schiske [14, 15], assuming a strong specular beam and a CTF-style model. Here the same FO forward map is inverted: given a through-focal intensity series $I(\mathbf{r}, \Delta z_k)$, $k = 1, \dots, K$, we recover ψ_0 within the aperture cutoff, up to a global phase.

A single defocus underdetermines a general complex object. The CTF factor may vanish on interior rings; the contrast aperture discards $|\mathbf{q}| > q_{\text{ap}}$; intensity is invariant under a global phase; and one complex measurement cannot identify the conjugate pair $(X(\mathbf{q}), X(-\mathbf{q}))$. A focal series with non-coincident zeros restores those modes that the aperture admits. We compare five inverses and recommend a two-stage estimator that uses the Yu 2019 kernel on the CTF slice $R_{\text{FO}}(\mathbf{q}, 0, \Delta z)$ and, when required, the full bilinear residual. Selected algebraic properties of the discrete linear stage are machine-checked in Lean 4 [23] against that kernel (Section 4 and Appendix A).

2 Image formation

The object is the complex exit wave

$$\psi_0(\mathbf{r}) = \sigma(\mathbf{r}) e^{i\phi(\mathbf{r})}, \quad (1)$$

with ϕ in radians. For a topographic height A in reflection the kinematic phase is

$$\phi = \frac{4\pi A}{\lambda_0}, \quad (2)$$

where λ_0 is the wavelength at the sample. At $E_0 = 10$ eV, $\lambda_0 \simeq 0.39$ nm, so $A = 0.1$ nm already gives $\phi \sim 3$, outside the weak-phase regime occupied by graphene ripples and atomic steps [10]. Lens optics use the wavelength λ after acceleration to column energy E (typically 15 kV): spatial frequency is $\mathbf{q} = \boldsymbol{\alpha}/\lambda$ [9, 8]. The reconstruction kernel below is written in λ ; conversion of reconstructed phase to height uses λ_0 .

With $\Psi = \mathcal{F}[\psi_0]$, the FO intensity is bilinear in the object spectrum [13, 7, 11],

$$I(\mathbf{r}, \Delta z) = \iint \Psi(\mathbf{q}) \Psi^*(\mathbf{q}') R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z) e^{2\pi i(\mathbf{q}-\mathbf{q}')\cdot\mathbf{r}} d^2q d^2q'. \quad (3)$$

The kernel is supported on the contrast aperture $|\mathbf{q}|, |\mathbf{q}'| \leq q_{\text{ap}}$ and factors as $R_{\text{FO}} = R_0 E_S E_{C,\text{tot}}$, with closed-form spatial and chromatic envelopes derived in [11] and proved in [27]. The CTF kernel agrees with R_{FO} on the slice $\mathbf{q}' = \mathbf{0}$,

$$R_{\text{FO}}(\mathbf{q}, \mathbf{0}, \Delta z) = R_{\text{CTF}}(\mathbf{q}, \Delta z). \quad (4)$$

The axis $\mathbf{q} = \mathbf{0}$ is the conjugate branch $R_{\text{FO}}(\mathbf{0}, -\mathbf{q})$, not the CTF. Wave aberration χ is measured in waves, with transfer factor $W = e^{2\pi i\chi}$ [9]. Defocus enters χ_S as $+\frac{1}{2}\Delta z\lambda q^2$ in the convention of this work; mixing the opposite sign used in some simulators interchanges overfocus and underfocus.

Intensity is invariant under $\psi_0 \mapsto e^{i\theta}\psi_0$. Modes with $|\mathbf{q}| > q_{\text{ap}}$ are identically invisible in every defocus. The linearization of (3) about a strong specular beam therefore recovers ψ_0 only up to a global phase (fixed here by taking the zero-frequency component real and positive) and only inside the aperture.

3 Through-focal inversion

Let N denote the number of image pixels and K the number of defoci; a typical stack is $N = 512^2$ and $K \sim 11$ –21. Regularization of an ill-posed filter follows Tikhonov [17] and Wiener [18]: a positive floor α in the denominator prevents division by a vanished CTF at the price of a bias of order α .

3.1 Candidate methods

Single-image Wiener deconvolution applies one Fourier transform and divides by $R_{\text{CTF}}(\mathbf{q}, \Delta z)$. Interior CTF zeros leave undetermined modes, and one plane does not identify a general complex object.

Transport of intensity [21, 22] recovers phase from two or three closely spaced defoci under a paraxial continuity equation. After acceleration, LEEM has $\lambda \sim 10$ pm, a hard aperture, spherical aberration and partial coherence. That PDE is not an inverse of (3) and has not been the method used on experimental LEEM graphene or steps.

Multi-defocus Schiske restoration [14, 12] transforms each plane and treats spatial frequencies independently. The slice multipliers are

$$h_k(\mathbf{q}) = R_{\text{FO}}(\mathbf{q}, \mathbf{0}, \Delta z_k). \quad (5)$$

For a single unknown amplitude $X(\mathbf{q})$ the regularized estimator is

$$\hat{x} = \frac{\sum_k \bar{h}_k y_k}{\alpha + \sum_k |h_k|^2}. \quad (6)$$

Linearization about vacuum couples $X(\mathbf{q})$ with $X(-\mathbf{q})$; the corresponding 2×2 regularized system uses the same α and is not (6). If one defocus has $h_k = 0$, the remaining planes can still determine the bin. The cost is $K+1$ discrete Fourier transforms plus $O(KN)$ arithmetic. This is Duden’s algorithm with the Yu 2019 envelopes, and it coincides with one Gauss–Newton step from vacuum on (3).

Gerchberg–Saxton iteration [19] and hybrid input–output [20] replace the computed modulus by the measured intensity and propagate with a coherent lens model. Under partial coherence the intensity constraint is the quadratic set of (3). Naive modulus replacement is therefore the coherent CTF model and is biased when FO and CTF disagree off axis [10]. Projection onto the FO intensity is a slower, undamped rewrite of nonlinear least squares.

Nonlinear least squares minimizes the bilinear FO residual of the stack. Each Gauss–Newton step uses the pair kernel, so it sees the mixing dropped by (6). That estimator is FO-faithful for $\phi \ll 1$, at the cost of pair accumulations rather than a Fourier-diagonal solve. If the aberrations themselves are unknown, the same idea stacked over extra defoci is the maximum-likelihood (MAL) reconstruction of Coene, Thust, Op de Beeck and Van Dyck [16].

Method	Typical cost	Fills CTF zeros	Faithful for large ϕ
Single-image Wiener	one transform	no	no
Transport of intensity	one transform	n/a (wrong PDE)	no
Multi-defocus regularized	$K+1$ transforms	yes ($K \geq 2$)	linearized only
Gerchberg–Saxton / HIO	many transforms	if $K \geq 2$	only if the projector is FO
Nonlinear least squares on FO	pair sums	yes	yes
Jacobi–Anger fit (1D sinusoid)	$O(M^2)$	n/a (parametric)	yes

3.2 Recommended estimator

For a general 2D experimental stack the fastest method that still fills CTF zeros is multi-defocus regularized inversion on the slice (4). It reads all KN Fourier samples, fills interior zeros of $\sin(2\pi\chi_S)$ by stacking defoci, and has a unique minimizer for every $\alpha > 0$ (Theorem A.1).

The FO-faithful estimator initializes that reconstruction and applies a few damped Gauss–Newton steps on the bilinear residual (MAL if aberrations are unknown). The nonlinear stage is omitted when the FO residual of the linear reconstruction already lies at the noise floor, equivalently $\max|\phi| \lesssim 0.3$ at typical LEEM signal-to-noise ($A \lesssim 0.01$ nm at $E_0 = 10$ eV). The linearization never sees off-axis pairs $(\mathbf{q}, \mathbf{q}')$ at which FO and CTF disagree; that omitted term is the quadratic remainder of (3) about vacuum (Theorem A.4) and is the whole image for a strong-phase sinusoid [10]. Agreement of the slice (4) with a CTF calculation is therefore not a sufficient skip criterion.

For the one-dimensional sinusoid of period Λ used in [10, 11] to exhibit CTF failure, the unknown reduces to an amplitude φ and a shift. The aperture retains $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1$ modes. The FO image is then a double sum of M^2 Jacobi–Anger terms,

$$e^{i\varphi \sin(2\pi x/\Lambda)} = \sum_{n \in \mathbb{Z}} J_n(\varphi) e^{2\pi i n x/\Lambda}, \quad (7)$$

and least squares on φ with $J'_n = \frac{1}{2}(J_{n-1} - J_{n+1})$ is $O(M^2)$. A general 2D micrograph has no Bessel ladder; the nonlinear stage remains Gauss–Newton in the aperture basis.

3.3 Discrete procedure

The linear stage is the unique minimizer of [Theorem A.1](#). After rigid alignment, each plane is mean-subtracted and discrete-Fourier transformed; the zero-frequency bin, which carries total flux ($R_{\text{FO}}(0,0) = 1$), is treated separately. For every $|\mathbf{q}| \leq q_{\text{ap}}$ and every recorded defocus the slice (5) is evaluated from the Yu 2019 envelopes. The scalar estimator (6) or the conjugate-pair 2×2 system is solved at that bin with fixed $\alpha > 0$; coefficients with $|\mathbf{q}| > q_{\text{ap}}$ are set to zero. An inverse transform yields $\sigma = |\hat{\psi}|$ and $\phi = \arg(\hat{\psi})$ in the vacuum gauge. Experimental lens current is mapped onto the defocus convention of [Section 2](#) by one fitted affine transform.

4 Machine-checked properties of the discrete estimator

Selected algebraic claims for the discrete linear estimator are machine-checked in Lean 4 [[23](#), [24](#)] against the FO kernel encoded from Yu *et al.* [[11](#), [27](#)]. The statements are finite-dimensional (finite defoci and a finite frequency grid). The discrepancy n_k that enters the error identity is deterministic (linearization remainder, instrument noise and discretization); no probability space is introduced. Full statements are collected in [Appendix A](#).

The reconstruction uses (4), not a substitute propagator. For $\alpha > 0$ the regularized energy at each frequency bin has a unique global minimizer (6) ([Theorem A.1](#)). If $y_k = h_k x^* + n_k$, the estimation error is exactly $(\sum_k \bar{h}_k n_k - \alpha x^*)/D$ with $D = \alpha + \sum_k |h_k|^2$ ([Theorem A.2](#)); a sharp triangle bound follows ([Theorem A.3](#)). Frequencies outside the aperture reconstruct as zero at every defocus; a global phase is invisible; one measurement does not identify $(X(\mathbf{q}), X(-\mathbf{q}))$; and $\sin(2\pi\chi_S)$ vanishes on interior rings of a pure-defocus column ($C_3 = C_5 = 0$, $\Delta z \neq 0$, large enough aperture), which is why $K \geq 2$ is required ([Theorem A.5](#)). The multi-defocus slice map is the derivative of bilinear FO at vacuum; the neglected term is homogeneous of degree two ([Theorem A.4](#)). When that remainder lies above the noise, the nonlinear stage is the consistent next estimator; the numerical skip criterion $\max|\phi| \lesssim 0.3$ is not part of the machine-checked fragment.

A modelled operation count, treating a discrete Fourier transform as $N \log_2 N$ operations, shows that the linear stage is cheaper than a dense pair apply already at $N = 128$. Existence of a fast Fourier transform, wall-clock times, Gaussian or Poisson statistics, Cramér–Rao bounds, Gauss–Newton convergence, image alignment and the current-to-defocus map are not claimed.

5 Conclusions

We invert the FO image-formation map of Yu, Lau and Altman [[11](#)] for a 2D through-focal intensity series. Regularized multi-defocus inversion on the CTF slice fills the zeros of a single defocus, is unique for every positive regularization, and is first-order consistent with bilinear FO. Gauss–Newton iteration on the full kernel is reserved for strong-phase objects whose linearized residual exceeds the noise; for the one-dimensional sinusoid that exhibits CTF failure it collapses to a Bessel fit. These results broaden quantitative LEEM contrast analysis from forward matching of a focal series to recovery of the object, with the discrete linear stage verified against the same kernel used to predict the images.

A Formal claims

The statements below are finite-dimensional. No existence theorem for the fast Fourier transform is claimed.

Theorem A.1 (Unique minimizer). *For $\alpha > 0$ the energy $J(x) = \sum_k \|h_k x - y_k\|^2 + \alpha \|x\|^2$ has a unique global minimizer (6). Completing the square: $J(x) = J(\hat{x}) + D \|x - \hat{x}\|^2$ with $D = \alpha + \sum |h_k|^2 > 0$.*

Lean: `tikhonovJ_min`, `tikhonov_unique`, `tikhonovJ_eq_shift`. The same identity gives uniqueness at $\alpha = 0$ whenever $\sum |h_k|^2 > 0$.

Theorem A.2 (Bias–noise identity). *If $y_k = h_k x^* + n_k$ and $D \neq 0$, then*

$$\hat{x} - x^* = \frac{\sum_k \overline{h_k} n_k - \alpha x^*}{D}. \quad (8)$$

Lean: `tikhonov_error`. Taking $\mathbb{E}[n] = 0$ is informal (no probability space).

Theorem A.3 (Triangle bound, sharp). *If $\alpha \geq 0$ and $D > 0$,*

$$|\hat{x} - x^*| \leq \frac{\sum_k |h_k| |n_k| + \alpha |x^*|}{D}. \quad (9)$$

Equality holds for one measurement, $h = n = 1$, $x^ = -1$, $\alpha = 1$.*

Lean: `tikhonov_error_bound`, `tikhonov_error_bound_sharp`.

Theorem A.4 (Quadratic remainder of bilinear FO). *With vacuum spectrum $1_{\mathbf{q}=0}$ and increment δ ,*

$$\hat{I}(\text{vac} + \delta) = \hat{I}(\text{vac}) + DI_{\text{vac}}[\delta] + \hat{I}(\delta), \quad (10)$$

where $DI_{\text{vac}}[\delta](\xi) = R_{\text{FO}}(\xi, 0) \delta(\xi) + R_{\text{FO}}(0, -\xi) \overline{\delta(-\xi)}$ and $|\hat{I}(\delta)(\xi)| \leq (\sum |R_{\text{FO}}|)(\sum |\delta|)^2$.

Lean: `ihat_add`, `ihatJac_vacuum`, `ihat_quadratic_remainder`, `ihat_bound`. The dropped term is exactly bilinear: linearized inversion is biased by $O(\|\delta\|^2)$, which is the whole image for a strong-phase sinusoid [10].

Theorem A.5 (Interior CTF zeros of a pure-defocus column). *$\sin(2\pi\chi_S) = 0$ whenever $2\chi_S \in \mathbb{Z}$. For $C_3 = C_5 = 0$, $\Delta z \neq 0$, and a large enough aperture, there exists nonzero q inside the disk with that vanishing (`exists_interior_weakPhase_zero`). A single defocus is therefore blind at those rings; $K \geq 2$ with incommensurate Δz typically fills them. This is a pure-defocus hypothesis, not a column that still has spherical aberration.*

The 2×2 Gram determinant of the vacuum Jacobian is positive for $\alpha > 0$ (`gramDet_pos`), so the conjugate-pair normal equations are invertible even though one measurement is not (Section 1).

Modelled cost. Set $\text{dftCost}(N) := N \log_2 N$ and

$$\text{reconstructCost}(K, N) = (K + 1) N \log_2 N + N(8K + 4). \quad (11)$$

For $K \geq 1$ and $N \geq 2$, $\text{reconstructCost}(K, N) \leq 14 K N \log_2 N$. At $N = 128$ this is strictly cheaper than a dense bilinear apply KN^2 , for every $K \geq 1$ (`reconstructCost_le`, `reconstructCost_lt_dense_128`). The constant 14 is attained in the model at $K = 1$, $N = 2$.

Claim	Lean
Tikhonov unique min, closed form	<code>LeemFO/Tikhonov.lean</code>
Bias–noise identity and sharp triangle bound	same
Cost model versus dense KN^2	<code>reconstructCost_*</code>
Aperture support of $R_{\text{FO}}(\cdot, 0)$	<code>LeemFO/LinearInverse.lean</code>
Gauge, vacuum Jacobian, quadratic remainder	same
$K = 1$ not injective; pure-defocus CTF zeros	same
Mode count $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1$	<code>LeemFO/Inverse.lean</code>
Jacobi–Anger and Bessel decay	<code>LeemFO/PhaseObject.lean</code>
FFT exists / equals the continuum FT	informal
Gaussian/Poisson noise, Cramér–Rao	informal
Gauss–Newton local quadratic convergence	informal
Image alignment, current-to-defocus map	informal
“Fastest among all conceivable algorithms”	only DFT versus dense bilinear

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